StudySapiens Solved Notes

Ch.4 Biomolecules

Biology 11 • Punjab English Medium • PECTAA 2025–26 • Solved Exercise
Prompts are paraphrased and keyed to textbook question numbers. MCQ answer letters refer to the textbook options.

SECTION 1: MULTIPLE CHOICE QUESTIONS

Answer letters refer to the textbook options. Full textbook stems/options are not reproduced here.

1. Which characteristic of water explains most of its unique properties?

Answer: (d) Water molecules stick together — Polarity and hydrogen bonding make water molecules cohere, giving high specific heat, heat of vaporization and solvent properties.

2. Human sex hormones belong to which lipid group?

Answer: (a) Steroids — Sex hormones such as testosterone and estrogen are steroid lipids with a four-ring structure.

3. Which of the following is not a protein?

Answer: (b) Cholesterol — Cholesterol is a steroid lipid; haemoglobin, pepsin and antibodies are proteins.

4. Which is the largest carbohydrate among the options?

Answer: (a) Cellulose — Cellulose is a polysaccharide of many glucose units; ribose, glyceraldehyde and glucose are small sugars.

5. Phosphorus shortage in soil would most directly reduce synthesis of which compound?

Answer: (a) DNA — DNA contains phosphate groups in its sugar-phosphate backbone; fatty acids, proteins and cellulose do not require phosphorus as a main component.

6. Which compound is most closely related in composition to maltose?

Answer: (a) Starch — Maltose is two glucose units; starch is a polymer of glucose.

7. Which group is found in all fatty acids?

Answer: (d) COOH — Fatty acids are long hydrocarbon chains with a terminal carboxyl group.

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8. Haemoglobin has which level of protein structure?

Answer: (d) Quaternary structure — Haemoglobin consists of multiple polypeptide subunits (2α and 2β chains).

9. Which process produces peptide bonds?

Answer: (b) Dehydration synthesis — A peptide bond forms by condensation between amino and carboxyl groups, releasing water.

SECTION 2: SHORT QUESTIONS

1. Sketch a hydrolysis reaction.

Hydrolysis

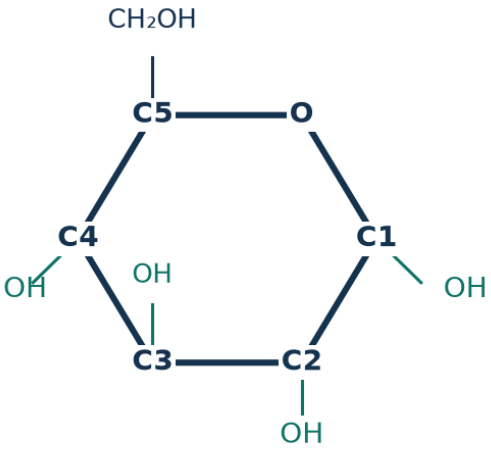
Polymer + H_2O → monomers

Example: maltose + H_2O → glucose + glucose

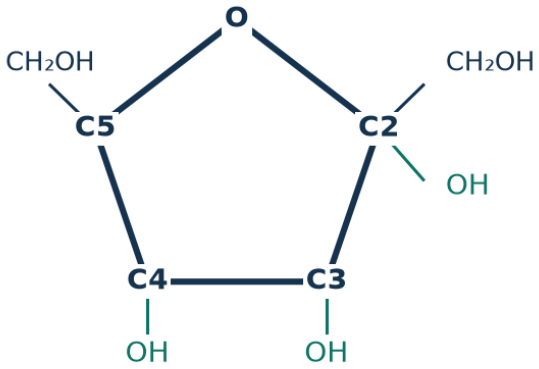
Water is added to break a covalent bond between monomers; digestive reactions are commonly hydrolytic.

2. Draw the ring structures of glucose and fructose.

α-D-glucose (six-membered ring)



Fructose (five-membered ring)



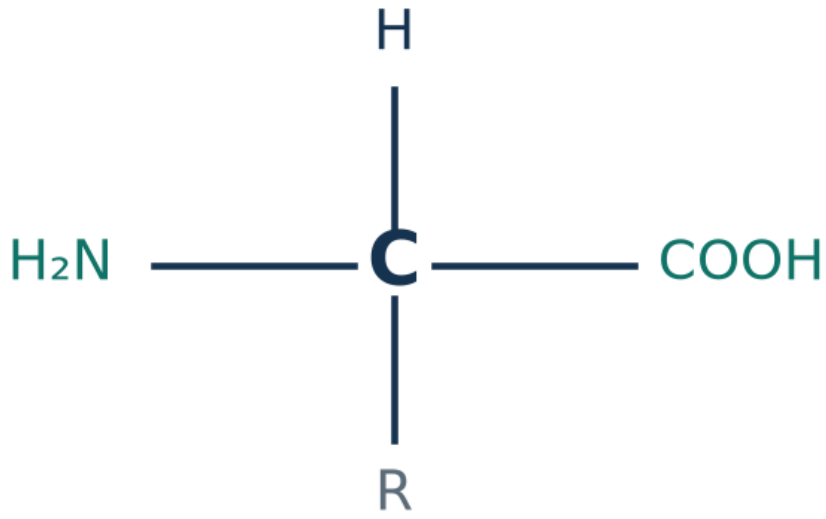
Ring forms: glucose forms a six-membered glucopyranose ring; fructose forms a five-membered fructofuranose ring.

In solution, glucose mainly forms a six-membered ring and fructose a five-membered ring. α- and β-forms differ in the orientation of the –OH at the anomeric carbon.

3. Define isomers and stereoisomers.

Isomers have the same molecular formula but different arrangements of atoms. Stereoisomers have the same bonding sequence but differ in spatial arrangement, e.g., glucose and galactose differ in the orientation of one –OH group.

4. Sketch an amino acid.



General amino acid (α-carbon with –NH_2, –COOH, –H and R group)

General amino acid: a central α-carbon bonded to –NH_2, –COOH, –H and a variable R group.

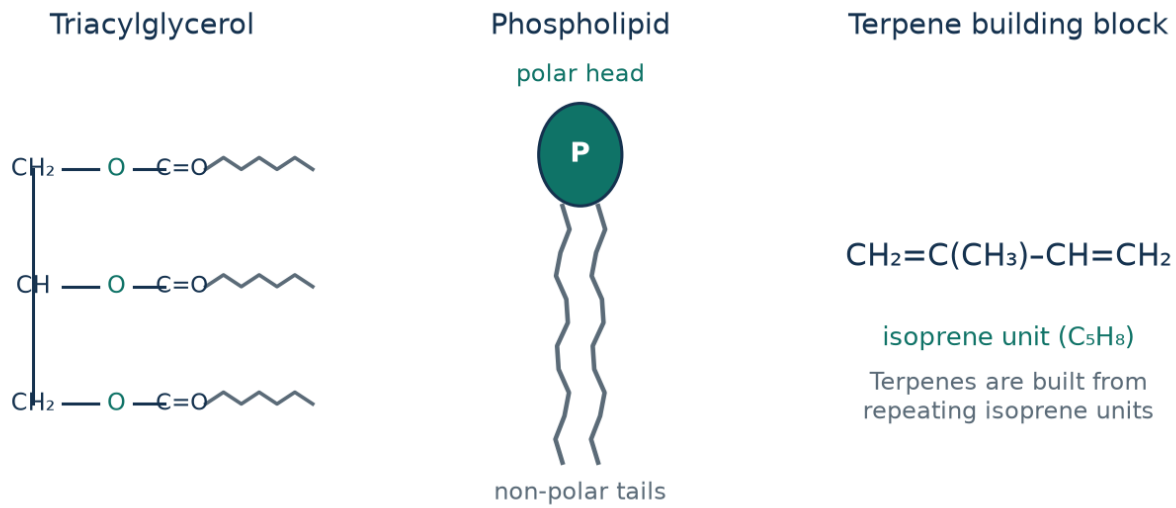
5. Outline the synthesis of peptide linkages.

Peptide bond

Amino acid 1 (–COOH) + amino acid 2 (–NH_2) → dipeptide (–CO–NH–) + H_2O

The carboxyl group of one amino acid reacts with the amino group of another by dehydration synthesis, forming a peptide bond.

6. Sketch acylglycerol, phospholipid and terpene.



Sketches: triacylglycerol (glycerol + 3 fatty acids), phospholipid (polar head + 2 non-polar tails) and the isoprene unit of terpenes.

7. Differentiate between nucleoside and nucleotide.

Nucleoside	Nucleotide
Components: Nitrogenous base + pentose sugar.	**Components:** Base + sugar + phosphate group(s).
Example: Adenosine.	**Example:** Adenosine monophosphate (AMP).
Role: Part of nucleic acid structure.	**Role:** Monomer of DNA/RNA; energy carriers such as ATP.

8. Illustrate formation of a phosphodiester bond.

Phosphodiester bond

3′–OH of one nucleotide sugar + 5′–phosphate of next nucleotide → phosphodiester linkage + H_2O

This creates the sugar-phosphate backbone of DNA/RNA with 5′→3′ directionality.

9. State the central dogma of gene expression.

DNA → RNA → Protein

Genetic information is transcribed from DNA to RNA and translated from RNA to protein.

SECTION 3: LONG QUESTIONS

1. Distinguish carbohydrates, proteins, lipids and nucleic acids as the four fundamental biomolecules.

Carbohydrates	Proteins	Lipids	Nucleic acids
Units: monosaccharides.	**Units:** amino acids.	**Units:** fatty acids/glycerol, isoprene units.	**Units:** nucleotides.
Bonds: glycosidic.	**Bonds:** peptide.	**Bonds:** ester (mostly).	**Bonds:** phosphodiester.
Elements: C, H, O (≈1:2:1).	**Elements:** C, H, O, N, often S.	**Elements:** C, H, O; little O, non-polar.	**Elements:** C, H, O, N, P.
Roles: energy source, storage, structure (cellulose).	**Roles:** enzymes, transport, structure, signalling, defence.	**Roles:** energy storage, membranes, hormones, insulation.	**Roles:** genetic information storage and expression.

Carbohydrates and lipids are mainly energy-related, proteins are functional/structural workhorses, and nucleic acids store and transmit genetic information.

2. Describe and sketch dehydration synthesis reactions.

Dehydration synthesis

Monomer + monomer → polymer + H_2O

Glucose + glucose → maltose + H_2O

Amino acid + amino acid → dipeptide + H_2O

Nucleotide + nucleotide → dinucleotide + H_2O

In dehydration (condensation) synthesis, an –OH is removed from one monomer and an –H from another, releasing water and forming a covalent bond. This builds polysaccharides (glycosidic bonds), proteins (peptide bonds), nucleic acids (phosphodiester bonds) and lipids (ester bonds). The reverse reaction, hydrolysis, adds water to break polymers.

3. Explain how the properties of water make it the medium of life.

- **Polarity and hydrogen bonding:** make water an excellent solvent for ions and polar molecules, providing a medium for metabolic reactions.
- **High specific heat:** absorbs/releases much heat with little temperature change, stabilizing internal and external environments.
- **High heat of vaporization:** evaporation removes heat, cooling organisms by sweating and transpiration.

- **Cohesion and adhesion:** support continuous water columns and transport in plant xylem; give surface tension.
- **Lower density of ice:** ice floats and insulates water below, protecting aquatic life.
- **Ionization:** H^+ and OH^- determine pH and participate in hydrolysis/condensation reactions.

These properties together make water indispensable as the medium of life.

4. Distinguish the properties and roles of monosaccharides and classify them.

Monosaccharides are simple sugars with empirical formula approximately $(CH_2O)n$. They are sweet, water-soluble and reducing. They are classified by carbon number:

Class	Carbon atoms	Examples
Trioses	3	Glyceraldehyde, dihydroxyacetone.
Tetroses	4	Erythrose, erythrulose.
Pentoses	5	Ribose, deoxyribose, ribulose.
Hexoses	6	Glucose, fructose, galactose.
Heptoses	7	Rare intermediates in photosynthesis.

Roles: immediate energy source (glucose), monomers for disaccharides/polysaccharides, and components of nucleic acids/ATP (ribose/deoxyribose).

5. Compare structural isomers and stereoisomers of glucose.

Structural isomers	Stereoisomers
Meaning: same molecular formula but different bonding/functional group.	**Meaning:** same bonding sequence but different spatial arrangement.
Example: glucose (aldose) and fructose (ketose), both $C_6H_{12}O_6$.	**Example:** glucose and galactose differ in –OH orientation at C-4.
Effect: different chemical properties and polymer formation.	**Effect:** α- and β-glucose give starch/glycogen versus cellulose.

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Thus small differences in structure or orientation produce large differences in biological properties.

6. Distinguish the properties and roles of disaccharides.

Maltose	Sucrose	Lactose
Monomers: glucose + glucose.	**Monomers:** glucose + fructose.	**Monomers:** glucose + galactose.
Bond: α-1,4-glycosidic.	**Bond:** α-1,2-glycosidic.	**Bond:** β-1,4-glycosidic.
Role/source: malt sugar; product of starch digestion.	**Role/source:** table sugar; transport sugar in phloem.	**Role/source:** milk sugar; energy for infant mammals.

Disaccharides form by dehydration synthesis and are broken by hydrolysis during digestion. They are less soluble than monosaccharides.

7. Define proteins and amino acids and outline synthesis/breakage of peptide linkages.

Proteins are polymers of amino acids joined by peptide bonds. An amino acid has an amino group ($–NH_2$), a carboxyl group (–COOH), a hydrogen atom and a variable R group attached to a central α-carbon.

Peptide linkage

$–COOH + –NH_2 \rightarrow –CO–NH– + H_2O$ (peptide bond)

Peptide + $H_2O \rightarrow$ amino acids (hydrolysis)

Synthesis: dehydration synthesis between the carboxyl group of one amino acid and the amino group of another releases water and forms a peptide bond. Breakage: hydrolysis adds water to break the bond, as in digestion or protein turnover. The sequence of amino acids determines protein structure and function.

8. Justify the significance of amino-acid sequence using sickle-cell haemoglobin.

The primary structure (amino-acid sequence) determines how a protein folds and functions. In sickle-cell anaemia, a single substitution in the β-globin chain—valine replaces glutamic acid at position 6—changes haemoglobin's surface properties.

Under low oxygen, HbS molecules polymerize into fibres, distorting red blood cells into sickle shapes. These cells are fragile, block capillaries and carry less oxygen. This single change shows that even one amino-acid substitution can alter protein structure and cause disease.

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9. Describe the properties and roles of acylglycerols, phospholipids, terpenes and waxes.

Acylglycerols	Phospholipids	Terpenes	Waxes
Structure: glycerol esterified with 1–3 fatty acids.	**Structure:** glycerol + 2 fatty acids + phosphate/polar head.	**Structure:** built from isoprene (C_5H_8) units.	**Structure:** long-chain fatty acid + long-chain alcohol.
Properties: hydrophobic; fats solid, oils liquid.	**Properties:** amphipathic; polar head, non-polar tails.	**Properties:** often hydrophobic; diverse pigments/oils.	**Properties:** water-repellent, high melting point.
Roles: energy storage, insulation, cushioning.	**Roles:** main component of cell membranes.	**Roles:** pigments, hormones, essential oils, vitamins (A, E, K).	**Roles:** protection and waterproofing in plants/animals.

10. Describe the molecular structure of a nucleotide.

- **Nitrogenous base:** purine (adenine, guanine; double ring) or pyrimidine (cytosine, thymine, uracil; single ring).
- **Pentose sugar:** ribose in RNA, deoxyribose in DNA.
- **Phosphate group(s):** attached to the 5′ carbon of the sugar.

The base attaches to the 1′ carbon of the sugar, forming a nucleoside; addition of phosphate makes a nucleotide. Nucleotides link by 3′–5′ phosphodiester bonds to form nucleic acids. ATP and NAD are important nucleotide derivatives.

11. Explain the Watson–Crick double-helix structure of DNA.

- Two antiparallel polynucleotide strands coil into a right-handed double helix.
- Sugar-phosphate backbones lie outside; nitrogenous bases face inward like ladder rungs.
- Adenine pairs with thymine by two hydrogen bonds; guanine pairs with cytosine by three hydrogen bonds.
- Each base pair contains one purine and one pyrimidine, keeping helix width constant.
- Strands are complementary, allowing accurate replication and transcription.
- The helix has major and minor grooves where regulatory proteins bind.

Chargaff's rules (A=T, G=C) and X-ray diffraction data supported this model.

12. Explain the general structure of RNA and differentiate the three types.

RNA is usually single-stranded, contains ribose and uracil instead of thymine, and can fold into secondary structures such as hairpins.

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mRNA	tRNA	rRNA
Structure: linear codon sequence.	**Structure:** cloverleaf/L-shaped; anticodon loop.	**Structure:** folded with proteins.
Function: carries genetic code from DNA to ribosome.	**Function:** brings specific amino acids to ribosome.	**Function:** structural and catalytic component of ribosomes.
Amount: about 3–4% of cellular RNA.	**Amount:** about 10–15% of cellular RNA.	**Amount:** about 80% of cellular RNA.

13. Define conjugated molecules and describe common examples.

Conjugated molecules are formed by combining two or more different classes of biomolecules.

- **Glycoproteins:** protein + carbohydrate; in membranes, secretions, cartilage and recognition.
- **Glycolipids:** lipid + carbohydrate; structural components of membranes.
- **Lipoproteins:** lipid + protein; transport lipids in blood and form membrane frameworks.
- **Nucleoproteins:** nucleic acid + protein; e.g., chromatin (DNA + histones) and ribosomes (rRNA + proteins).

These combinations give molecules properties that neither component has alone.

INQUISITIVE QUESTIONS

1. What happens if one amino acid is substituted in a polypeptide? Give an example.

A substitution can alter folding, charge, active-site shape or stability, changing function. Example: sickle-cell haemoglobin, where valine replaces glutamic acid in β-globin, causing HbS polymerization and sickled red cells.

2. How does a protein’s three-dimensional structure relate to its function?

Function depends on shape: active sites, binding pockets and interaction surfaces are formed by folding. Denaturation disrupts this shape and usually destroys function, showing that structure determines activity.

3. How do nucleic acids encode information and how is it translated into proteins?

The sequence of bases in DNA stores genetic information. Transcription produces mRNA; ribosomes read mRNA codons, and tRNA brings matching amino acids. The amino-acid sequence folds into a functional protein.

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